Discrete Structures: Basic Structures – Part 1: Sets

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Introduction 1

- A logistics company is building a new automatic sorting system for packages.
- Each day, the warehouse receives a lot of packages, where each package has a unique identifier. Each package is assigned to one and only one truck (truck 1-4) based on its ZIP code.
  - As the conveyor runs, packages are scanned in order of arrival, and assigned to the corresponding truck.
  - Each truck's total load also needs to be monitored, to ensure it doesn't go beyond the limit.

Introduction 2

- Each package has a unique identifier
- Packages are scanned in order of arrival
- Each package is assigned to extract one truck
- Each truck's total load stays within the capacity

A set of unique IDs:
P = {pkg_1, pkg_2, …}

There can be a sequence of packages:
$a_1, a_2, a_3, \ldots$

It can be defined as a function
$f: P \rightarrow \{t1, t2, t3, t4\}$,
i.e.,
$f(a_1), f(a_2), f(a_3), \ldots$

The summation of all packages on the truck should be calculated:
$Load_2 = \Sigma_{1 \leq i \leq n} g_i[f(a_i) = t2] \leq Capacity_2$

Outline For This Chapter and Lecture

- Basic Structures
  - Set
    - Applications
    - Definitions
    - Size / cardinality
    - Representations
    - Equivalence
    - (Proper) subset
    - Set operations
  - Function
  - Sequence
  - Summation

Set in Real Life

- How to get the collection of our last names without duplication?
  - Remove the duplicated names in the array
  - How to detect duplicated names?
    - Sort?
    - Compare with checked names? How to save checked ones?
  - Can we make it more efficient?
- In database, primary keys are used as identifiers and enforce uniqueness
- In search engines, we need to have a unique reference / URL to each website
- In a computer network, we need to have the unique IPs of active machines
- In machine learning, we can use a set of features to identify objects

Set Definition

- Textbook: A *set* is an unordered collection of objects.
- C++: Sets are containers that store unique elements following a specific order.
  http://www.cplusplus.com/reference/set/set/
  http://www.cplusplus.com/reference/unordered_set/unordered_set/
- Java/Python: A collection that contains no duplicate elements.
  https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/Set.html
  https://docs.python.org/3/tutorial/datastructures.html#sets
- In our class: An **unordered** collection of **unique** objects.
  - Realize the difference between theory (math in textbook) and real practices (support in programming languages)

Examples of Sets

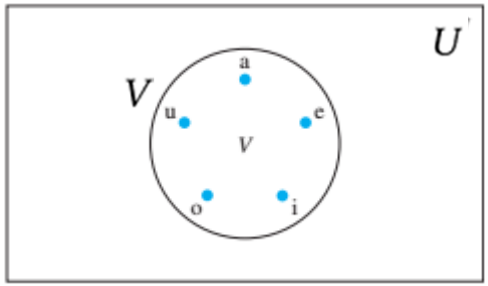
- **Example**: the collection of integers whose values are between 1 and 99
  - $I = \{1,2,3,\cdots,99\}$.
- **Example**: The set *V* of all vowels in the English alphabet $V = \{a, e, i, o, u\}$.
- **Empty/null set**: if a set has no elements, we call it an empty set or a null set, and it is denoted by $\emptyset$.
- A set is said to ***contain*** its elements
  - Notation: We use $\boldsymbol{a \in A}$ to describe a is an element of set A. And use $a \notin A$ to describe a is NOT an element of set A.

Size of Sets

- Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and that n is the ***size*** or ***cardinality*** of S. The cardinality of S is also denoted by $|S|$.
  - Example: Let A be the set of odd positive integers less than 10. $A = \{1,3,5,7,9\}$ and $|A| = 5$.
  - Example: Because the **null set** has no elements, it follows that $|\emptyset| = \mathbf{0}$.
  - Example: The set of positive integers is **infinite**.
    - The basic representation like {1, 2, 3, …} is not good enough – we need better representations.

Represent a Set

- Roster method
  - List all the unique elements inside a pair of braces, e.g., {1, 2, 3, 4, …, 99}
- Set Builder Notation
  - Specify the domain, format, and properties of the elements, e.g., {x | x is a positive integer less than 100}, which means “the set of all x such that x has property ‘x is a positive integer less than 100’”.
- Venn Diagram
  - Use a visualized graph to roughly illustrate the key idea of the set



Set Builder Notations 1

- Set Builder Notation
  - Specify the domain, format and properties of the elements
- How to quickly specify the rough domain in convention, e.g. the range for integers / real numbers, etc.

$\mathbb{N}$	$\mathbb{N} = \{0,1,2,\cdots\}$, **the set of natural numbers**
$\mathbb{Z}$	$\mathbb{Z} = \{\cdots,-2,-1,0,1,2,\cdots\}$, the set of integers
$\mathbb{Z}^+$	$\mathbb{Z}^+ = \{1,2,3,\cdots\}$, the set of positive integers
$\mathbb{Q}$	$\mathbb{Q} = \{{}^{p}/_{q} \mid \mathbf{p} \in \mathbb{Z}, \mathbf{q} \in \mathbb{Z}, \mathbf{q} \neq \mathbf{0}\}$, the set of rational numbers
$\mathbb{R}$	the set of real numbers

This is similar as data type, with potential sign restriction.

Set Builder Notations 2

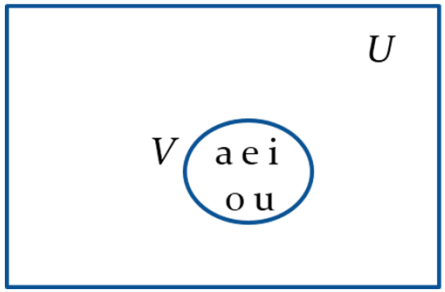
- Examples:

Set builder notation	Explanation
$\boldsymbol{A} = \{x \mid x$ **is odd positive integer less than 10**$\}$.	$A = \{1,3,5,7,9\}$.
$\boldsymbol{B} = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$.	B is the set of positive integers given that x is odd and x is less than 10. $B = \{1,3,5,7,9\}$.
$\mathbb{Q}^+ = \{x \in \mathbb{R} \mid x = {}^{p}/_{q}, p \in \mathbb{Z}^+, q \in \mathbb{Z}^+\}$	$\mathbb{Q}^+$ is the set of all positive rational numbers.
$\boldsymbol{C} = \{(\boldsymbol{2k+1}, \boldsymbol{2j}) \mid \boldsymbol{k} \in \mathbb{Z}, \boldsymbol{j} \in \mathbb{Z},\}$	$\boldsymbol{C}$ is the set of **pairs of integers**, and in each pair one integer is odd and the other is even

Try to use pure math notations.

Venn Diagram

- Venn Diagram is commonly used to visualize the sets of interest in the same graph
  - The entire domain is named as the ***universal set*** U, represented by a **rectangle**.
    - Universal set could be different in different cases
      - Example: could be all integers, or all real numbers.
  - Inside the rectangle, **circles**, **ellipse** or other geometrical figures are used to represent sets.
  - Typically used to **illustrate the relations between sets**.
  - Example: Draw a Venn diagram that represents *V,* the set of vowels in the English alphabet.



Equivalent Sets

- **Two sets are equal** if and only if they have the same elements.
  - That is if *A* and *B* are sets, then ***A* and *B* are equal iff** $\forall \boldsymbol{x}(\boldsymbol{x} \in \boldsymbol{A} \leftrightarrow \boldsymbol{x} \in \boldsymbol{B})$.
  - We write $\boldsymbol{A} = \boldsymbol{B}$ if *A* and *B* are equal sets.
    - Of course, if $A = B$ then $|A| = |B|$
  - E.g. The sets $\{1,3,5\}$ and $\{3,5,1\}$ are equal
    - $|\{1,3,5\}| = |\{3,5,1\}| = 3$
  - E.g. $\{1,3,5\}$ and $\{1,3,5,3,3,3,3,1\}$ are equivalent sets.
    - $|\{1,3,5\}| = |\{1,3,5,3,3,3,3,1\}| = 3$
    - In a computer, duplicated elements would waste our memory if we only count duplicate once, so practically we don't have duplicated elements in a set.

Subset

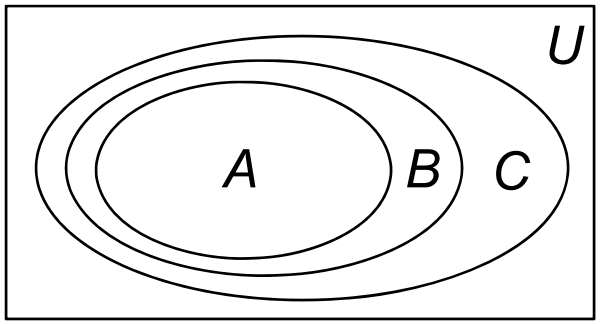
- The set A is said to be a **subset** of B if and only if every element of A is also an element of B.
  - We use the notation $\boldsymbol{A \subseteq B}$ to indicate that A is a subset of the set B
  - We say $A \subseteq B$ iff $\boldsymbol{\forall x(x \in A \rightarrow x \in B)}$
  - For every set S, $\emptyset \subseteq S$ and $S \subseteq S$
  - For two sets S and T, if $S \subseteq T$ then $|S| \leq |T|$
  - Example:
    - $\emptyset \subseteq \{1,2,3,4,5\}$
    - $\{3,4\} \subseteq \{1,2,3,4,5\}$
    - $\{1,2,3,4,5\} \subseteq \{1,2,3,4,5\}$
  - $\boldsymbol{A = B \leftrightarrow A \subseteq B \wedge B \subseteq A}$
    - Think about: for integers a and b, a ≤ b ∧ b ≤ a ↔ a = b

Proper Subset

- Set A is a **proper subset** of B when $A \subseteq B$ and there is at least one element in B that is not in A. We use the notation $\boldsymbol{A \subset B}$ when A is a proper subset of B.
- We say $A \subset B$ iff

$$\forall \boldsymbol{x}(\boldsymbol{x} \in \boldsymbol{A} \rightarrow \boldsymbol{x} \in \boldsymbol{B}) \wedge \exists \boldsymbol{x}(\boldsymbol{x} \in \boldsymbol{B} \wedge \boldsymbol{x} \notin \boldsymbol{A})$$

  - **Example**: Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.
    - Smaller ellipse inside another



  - For two sets S and T, if $S \subset T$ then $|S| < |T|$

Set Operations

- Like other objects/structures, after defining the states, we need to define the behaviors. Cartesian Product
  - Relation
  - Complement
  - Union
  - Intersection
  - Difference
- Set Identities
  - Common rules of set operations

Cartesian Product

- ***Cartesian Product*** is the basis for ***relations***
  - This is the foundation for relational database
- Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of **all ordered tuples $(\boldsymbol{a}, \boldsymbol{b})$, where $\boldsymbol{a} \in \boldsymbol{A}$ and $\boldsymbol{b} \in \boldsymbol{B}$.**
  - Hence, $\boldsymbol{A} \times \boldsymbol{B} = \{(\boldsymbol{a}, \boldsymbol{b}) | \boldsymbol{a} \in \boldsymbol{A}, \boldsymbol{b} \in \boldsymbol{B}\}$.
  - A **binary relation from $\boldsymbol{A}$ to $\boldsymbol{B}$** is a subset of $A \times B$
  - Ordered tuple: elements in the tuple are ordered
  - **Example**: the Cartesian product of $A = \{1,2\}$ and $B = \{a, b, c\}$ is
    $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$.

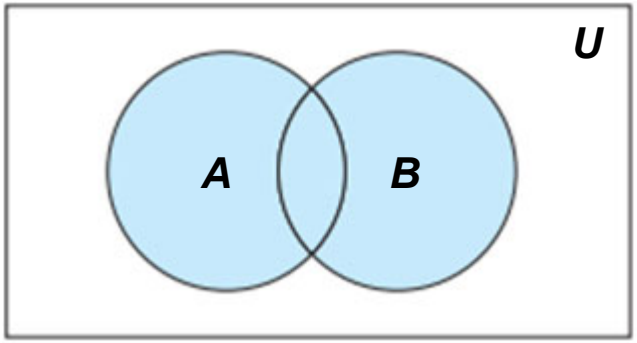
Union and Intersection of Sets

- Let *A* and *B* be sets. The ***union*** of the sets *A* and *B*, denoted by $\boldsymbol{A \cup B}$, is the set that contains those elements that are **either in *A* or in *B***
  - $\boldsymbol{A \cup B = \{x | x \in A \vee x \in B\}}$
  - Example: if $A = \{1,3,5\}$ and $B = \{1,2,3\}$, then
    $$A \cup B = \{1,3,5\} \cup \{1,2,3\} = \{1,2,3,5\}$$
- The ***intersection*** of the sets *A* and *B*, denoted by $\boldsymbol{A \cap B}$, is the set that contains those elements that are **in both *A* and *B***:
  - $\boldsymbol{A \cap B = \{x | x \in A \wedge x \in B\}}$
  - Example: if $A = \{1,3,5\}$ and $B = \{1,2,3\}$, then
    $$A \cap B = \{1,3,5\} \cap \{1,2,3\} = \{1,3\}$$

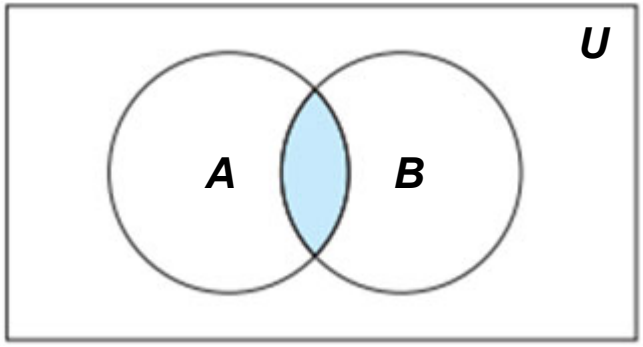
Complement of Sets

- Let *A* and *B* be sets. The ***difference*** of *A* and *B*, denoted by $\boldsymbol{A - B}$, is the set containing those elements that are **in *A* but not in *B***.
  - The difference of *A* and *B* is also called the ***complement of B with respect to A***
  - $\boldsymbol{A - B = \{x | x \in A \wedge x \notin B\}}$
  - Example: if $A = \{1,3,5\}$ and $B = \{1,2,3\}$, then $A - B = \{5\}$
- Let *U* be the universal set. The ***complement*** of the set *A*, denoted by $\overline{\boldsymbol{A}}$, is the complement of *A* respect to *U*.
  - The complement of the set *A* is $\boldsymbol{U - A}$
    - $\overline{\boldsymbol{A}} = \{\boldsymbol{x} | \boldsymbol{x} \notin \boldsymbol{A}\}$ **or** $\overline{\boldsymbol{A}} = \{\boldsymbol{x} | \neg(\boldsymbol{x} \epsilon \boldsymbol{A})\}$
  - Example: if $A = \{a, e, i, o, u\}$ and the ***universal*** set is the set of letters of the English alphabet, then
    $\overline{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, , q, r, s, t, v, w, x, y, z\}$

Venn Diagram for Set Operations

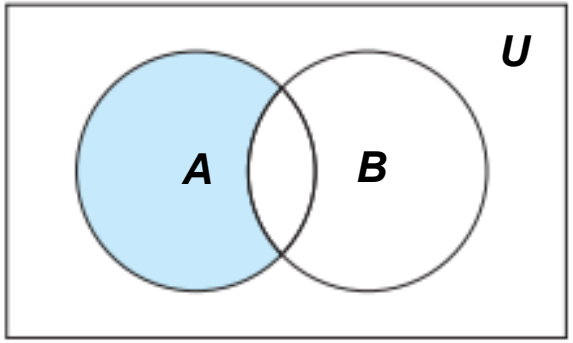


$A \cup B$ is shaded

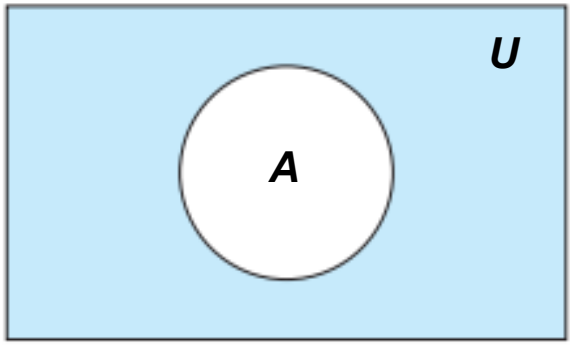


$A \cap B$ is shaded

$$|A \cup B| = |A| + |B| - |A \cap B|$$



$A - B$ is shaded



$\overline{A}$ is shaded

$$A - B = A \cap \overline{B}$$

Set Identities 1

- Basic identities from common sense
  - These rules are closely related to propositional rules
    - $A \cup B = \{x | x \in A \vee x \in B\}$
    - $A \cap B = \{x | x \in A \wedge x \in B\}$
    - $\bar{A} = \{x | \neg(x \epsilon A)\}$
    - If we model $x \epsilon A$ as $A(x)$ and $x \epsilon B$ as $B(x)$,
      - $A \cup B$ is very similar as $A(x) \vee B(x)$
      - $A \cap B$ is very similar as $A(x) \wedge B(x)$
      - $\bar{A}$ is very similar as $\neg A(x)$

Identity	Name	Identity	Name
$A \cup \emptyset = A$ $A \cap U = A$	Identity laws	$\overline{(\bar{A})} = A$	Complementation law
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws	$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws	$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

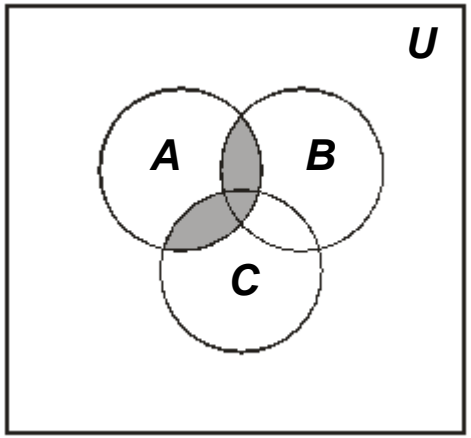
Set Identities 2

- Set identities can be proved by **membership table**
  - We categorize elements into all different cases, and verify whether both sets contain the elements in the same cases
  - 1 means elements in a set, 0 means elements not in a set
- Let *A*, *B*, and *C* be sets, show that
  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Set Identities 3

- Using the Venn Diagram, we can also verify one of the $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
  - $(A \cap B) \cup (A \cap C)$
  - $A \cap (B \cup C)$



Practice 1

- 1. Use set builder notation to give a description of each set below.
  - {0, 3, 6, 9, 12}
  - {-3, -2, -1, 0, 1, 2, 3}
  - {-3, -1, 1, 3, 5, 7}

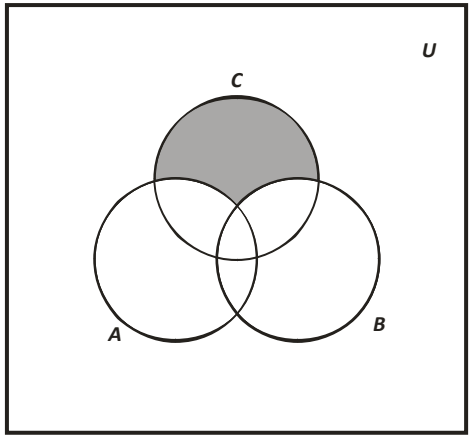
Practice 2

- 1. Use set builder notation to give a description of each set below.

  - $\{0, 3, 6, 9, 12\}$
    - $\{ 3n \mid n = 0, 1, 2, 3, 4\}$ or $\{ 3n \mid n \in \mathbb{N}, n \leq 3\}$
      $\{ x \mid x \text{ is a multiple of } 3 \wedge 0 \leq x \leq 3\}$
  - $\{-3, -2, -1, 0, 1, 2, 3\}$
    - $\{ x \in \mathbb{Z} \mid -3 \leq x \leq 3 \}$
  - $\{-3, -1, 1, 3, 5, 7\}$
    - $\{ 2k + 1 \mid k \in \mathbb{Z}, -2 \leq k \leq 3\}$

Practice 3

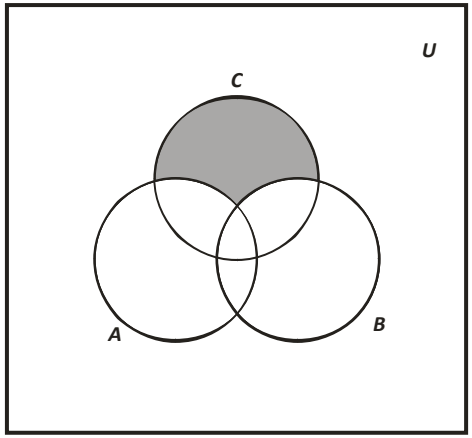
- 2. Give a set that can describe the following Venn diagram



Practice 4

- 2. Give a set that can describe the following Venn diagram

  - $C \cap \overline{(A \cup B)}$
  - $C - \mathrm{A} - \mathrm{B}$



Practice 5

- 3. Let A={0,2,4}, B={0,1,2,3,4}, and C={4, 5, 6, 7, 8}. Find
  - $C - B$
  - $(A \cup B) \cap C$
  - $(A \cap B) \cup C$
  - $A \times B$

Practice 6

- 3. Let A={0,2,4}, B={0,1,2,3,4}, and C={4, 5, 6, 7, 8}. Find
  - C - B
    - {5, 6, 7, 8}
  - $(A \cup B) \cap C$
    - {4}
  - $(A \cap B) \cup C$
    - {0,2,4,5,6,7,8}
  - $A \times B$
    - {(0,0), (0,1), (0,2), (0,3), (0,4), (2,0), (2,1), (2,2), (2,3), (2,4), (4,0), (4,1), (4,2), (4,3), (4,4) }

If $A \cup B = B$, do you think $A \subseteq B$ will always hold?

If $A \cap B = A$, do you think $A \subseteq B$ will always hold?

Extension 1

- 1. About empty set
  - Please note $\emptyset$ **itself is a set**
  - $\emptyset \neq \{\emptyset\}$
    - $\{\emptyset\}$ is a set containing an empty set. $|\emptyset| = \mathbf{0}$, $|\{\emptyset\}| = \mathbf{1}$.
  - $|\{\emptyset, \{\emptyset\}\}| = \mathbf{2}$
    - There are two elements in this set: $\emptyset$ and $\{\emptyset\}$
  - $\emptyset \in \{\emptyset, \{\emptyset\}\}$, $\emptyset \subseteq \{\emptyset, \{\emptyset\}\}$, $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$
    - $\emptyset \in \{\emptyset, \{\emptyset\}\}$ holds because $\{\emptyset, \{\emptyset\}\}$ contains two elements: $\emptyset$ and $\{\emptyset\}$, so $\emptyset$ is contained in $\{\emptyset, \{\emptyset\}\}$
    - $\emptyset \subseteq \{\emptyset, \{\emptyset\}\}$ holds because $\emptyset$ is a subset of any set
    - $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$ holds because all element in $\{\emptyset\}$ are also in $\{\emptyset, \{\emptyset\}\}$
    - More: $\{\emptyset\} \nsubseteq \{\{\emptyset\}\}$ but $\{\emptyset\} \in \{\{\emptyset\}\}$

Extension 2

- 2. Find sets A and B such that $A \subseteq B$ and $A \in B$
  - We can use examples in the previous page as one possible solution
    - $\boldsymbol{A} = \emptyset$
    - $\boldsymbol{B} = \{\emptyset, \{\emptyset\}\}$
    - $\emptyset \in \{\emptyset, \{\emptyset\}\},\ \emptyset \subseteq \{\emptyset, \{\emptyset\}\}$
  - Even $\{A\} \subseteq B$ and $\{A\} \in B$ hold
    - $\{\emptyset\} \in \{\emptyset, \{\emptyset\}\},\ \{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$

Extension 3

- 3. Set identities vs. equivalence in propositions

Set Identity	Name
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws

Other Logical Equivalences	
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws

Extension 4

- Set identities are closely related to propositional rules
  - Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Steps	Set Notations	Explanations
1	$\overline{A \cap B}$	Left hand side
2	$\{x \mid x \notin A \cap B\}$	Definition of complement
3	$\{x \mid \neg\big((x \in A) \wedge (x \in B)\big)\}$	Definition of intersection and negation
4	$\{x \mid \neg(x \in A) \vee \neg(x \in B)\}$	De Morgan's law from logic
5	$\{x \mid (x \notin A) \vee (x \notin B)\}$	Definition of negation
6	$\{x \mid \big(x \in \overline{A}\big) \vee (x \in \overline{B})\}$	Definition of complement
7	$\{x \mid x \in \overline{A} \cup \overline{B}\}$	Definition of union
8	$\overline{A} \cup \overline{B}$	Right hand side

Extension 5

- Interview practice
  - https://leetcode.com/problems/single-number-iii/description/